

Sinangoe vs. Zabalo

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Sinangoe versus Zabalo

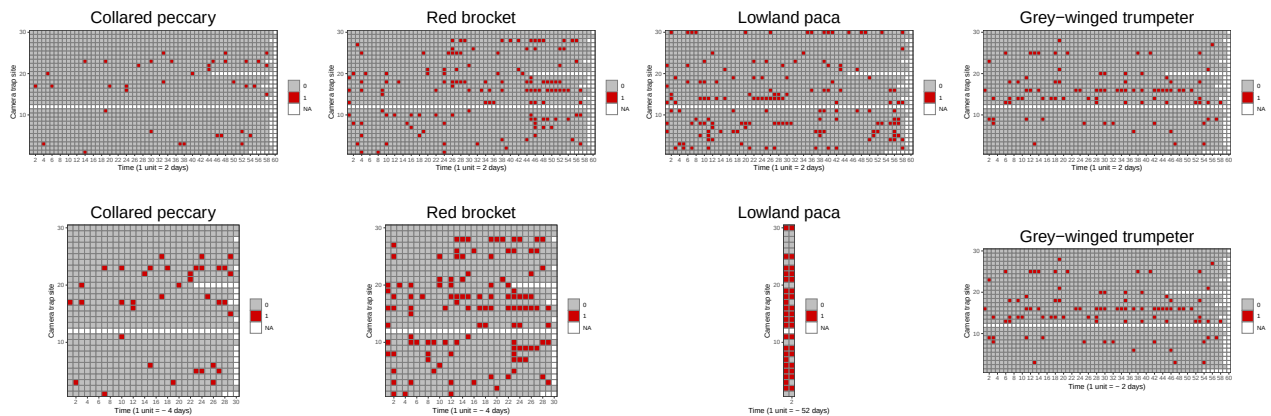
Timeframe

In Zabalo, camera trapping commenced on 2018-07-03 and ended on 2018-10-09, spanning 98 days. As for Sinangoe, trapping began on 2022-08-02 and ended on 2022-12-01, going for 121 days.

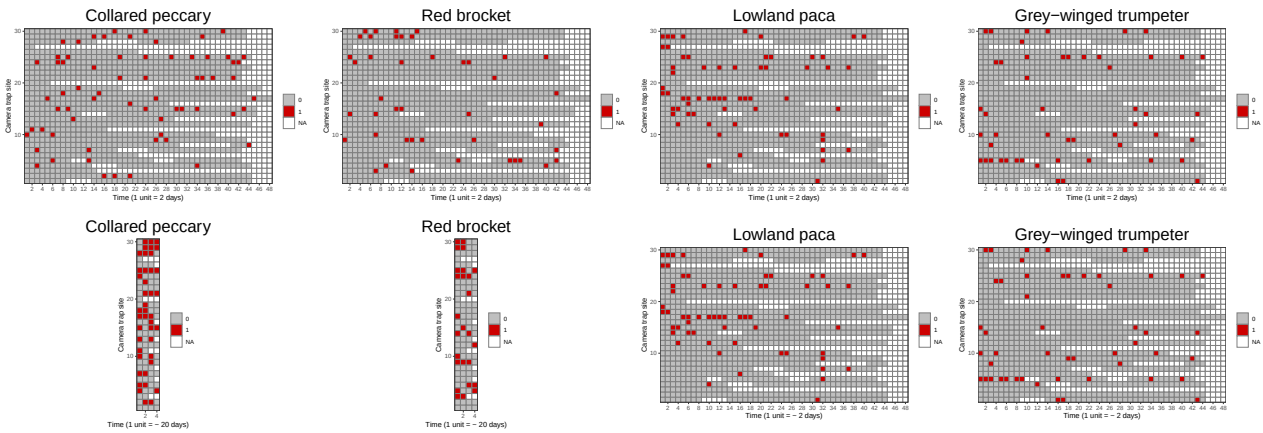
Consolidating time steps

The following plots show unconsolidated versus consolidated time steps for three species in Sinangoe and in Zabalo.

Sinangoe



Zabalo



Top species

Zabalo

Table 1: Zabalo Species

	Species	Total
4	Cuniculus paca	79
7	Dicotyles tajacu	78
14	Mazama gouazoubira	72
25	Psophia crepitans	59
5	Dasyprocta fuliginosa	56
13	Mazama americana	48
30	Tayassu pecari	42
1	Atelocynus microtis	40
15	Mitu salvini	32
29	Tapirus terrestris	32
6	Dasypus novemcinctus	22
18	NA NA	20
9	Eira barbara	15
3	Crypturellus undulatus	12
17	Myrmecophaga tridactyla	12

Sinangoe

Table 2: Sinangoe Species

	Species	Total
23	N/D N/D	854
20	Metachirus nudicaudatus	371
8	Dasyprocta fuliginosa	312
7	Cuniculus paca	171
19	Mazama americana	156
28	Philander andersoni	136
9	Dasypus novemcinctus	135
13	Geotrygon montana	132
39	Tinamus major	110
11	Didelphis marsupialis	107
34	Psophia crepitans	103
24	Nasua nasua	52
25	Nothocrax urumutum	52
10	Dicotyles tajacu	45
15	Hadrosaurus spadiceus	39

Occupancy modeling

Proceeding with all models for Sinangoe and models without missing estimates or disproportionately large standard errors for Zabalo.

Zabalo

Table 3: Collared peccary

	ModelName	AIC	diffFromBest
7	~1 ~ 1	140.4282	0.000000
2	~1 ~ Trail.Distance	142.1786	1.750435
3	~1 ~ HuntingIntensity	142.2508	1.822676

Table 4: Red brocket

	ModelName	AIC	diffFromBest
4	~Effort + Habitat ~ 1	112.9465	0.000000
1	~Effort + Habitat ~ Trail.Distance	114.0893	1.142761

	ModelName	AIC	diffFromBest
2	~Effort + Habitat ~ HuntingIntensity	114.4834	1.536911

Table 5: Lowland paca

	ModelName	AIC	diffFromBest
8	~Effort + Habitat ~ 1	466.9433	0.000000
2	~Effort + Habitat ~ Trail.Distance	468.1459	1.202603
5	~Effort + Habitat ~ Habitat + HuntingIntensity	468.5897	1.646409
3	~Effort + Habitat ~ HuntingIntensity	468.8361	1.892793

Table 6: Grey-winged trumpeter

	ModelName	AIC	diffFromBest
2	~Effort ~ Trail.Distance + HuntingIntensity	379.2717	0.000000
1	~Effort ~ HuntingIntensity	380.6071	1.335412

Top Models

Collared peccary

```
## Occupancy:
## Estimate SE z P(>|z|)
## 1.26 0.596 2.12 0.0343
##
## Detection:
## Estimate SE z P(>|z|)
## 0.00419 0.261 0.0161 0.987
```

Red brocket

```
## Occupancy:
## Estimate SE z P(>|z|)
## 0.835 0.647 1.29 0.197
##
## Detection:
## Estimate SE z P(>|z|)
## (Intercept) 0.0981 0.899 0.109 0.9131
## Effort 0.8442 0.361 2.340 0.0193
## HabitatSwamp 0.5960 1.186 0.502 0.6154
## HabitatUpland -1.3321 1.000 -1.332 0.1827
```

Lowland paca

```
## Occupancy:
## Estimate SE z P(>|z|)
## 1.3 0.56 2.32 0.0205
##
## Detection:
## Estimate SE z P(>|z|)
## (Intercept) -1.840 0.234 -7.863 3.74e-15
## Effort 0.103 0.178 0.580 5.62e-01
## HabitatSwamp -0.680 0.828 -0.821 4.12e-01
## HabitatUpland -0.933 0.312 -2.993 2.76e-03
```

Grey-winged trumpeter

```
## Occupancy:
## Estimate SE z P(>|z|)
## (Intercept) 0.764 0.641 1.19 0.2332
## Trail.Distance 1.170 0.711 1.65 0.0997
## HuntingIntensity 2.354 1.184 1.99 0.0468
##
## Detection:
## Estimate SE z P(>|z|)
## (Intercept) -2.39 0.154 -15.48 4.48e-54
## Effort -0.47 0.186 -2.53 1.13e-02
```

Sinangoe

Table 7: Collared peccary

	ModelName	AIC	diffFromBest
2	~1 ~ 1	289.1430	0.000000
1	~1 ~ DistToComm	291.0196	1.876652

Table 8: Red brocket

	ModelName	AIC	diffFromBest
2	~1 ~ 1	667.8246	0.000000
1	~1 ~ DistToComm	669.4884	1.663769

Table 9: Lowland paca

	ModelName	AIC	diffFromBest
2	~1 ~ 1	59.54592	0.000000
1	~1 ~ DistToComm	61.43766	1.891741

ModelName	AIC	diffFromBest
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Table 10: Grey-winged trumpeter

ModelName	AIC	diffFromBest
~1 ~ DistToComm	606.6556	0

Top Models

Collared peccary

```
## Occupancy:
## Estimate SE z P(>|z|)
## 0.0799 0.408 0.196 0.845
##
## Detection:
## Estimate SE z P(>|z|)
## -2.37 0.194 -12.2 3.6e-34
```

Red brocket

```
## Occupancy:
## Estimate SE z P(>|z|)
## 1.87 0.558 3.36 0.000791
##
## Detection:
## Estimate SE z P(>|z|)
## -1.65 0.103 -16.1 1.49e-58
```

Lowland paca

```
## Occupancy:
## Estimate SE z P(>|z|)
## 0.987 0.423 2.33 0.0196
##
## Detection:
## Estimate SE z P(>|z|)
## 2.48 0.624 3.98 6.76e-05
```

Grey-winged trumpeter

```
## Occupancy:
## Estimate SE z P(>|z|)
## (Intercept) 0.174 0.429 0.405 0.6853
```

```
## DistToComm      1.322 0.622 2.125  0.0335
##
## Detection:
## Estimate      SE      z  P(>|z|)
##      -2.17 0.113 -19.2 4.87e-82
```

Diversity and richness

Species richness Zabalo and Sinangoe had 30 and 38 total species, respectively, with Menhinick's species richness indices of 1.127 and 0.696.

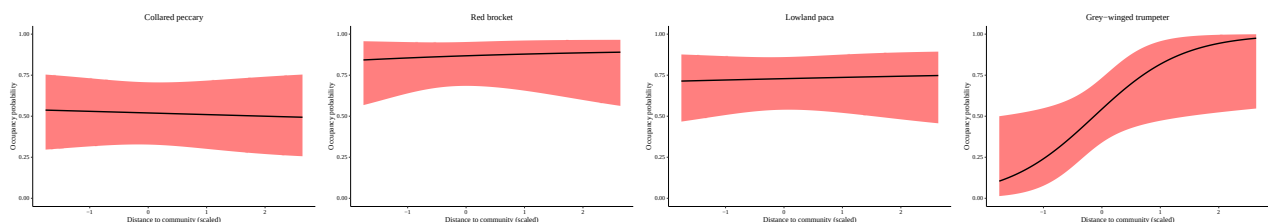
Diversity Among Sinangoe stations, the Shannon Diversity Index ranged from 0.305 to 2.589. Among Zabalo stations, the Shannon Diversity Index ranged from 0 to 2.73. Among Siona stations, the Shannon Diversity Index ranged from 0 to 2.548. Among Siekopai stations, the Shannon Diversity Index ranged from 0.349 to 2.269.

Table 11: Species Diversity

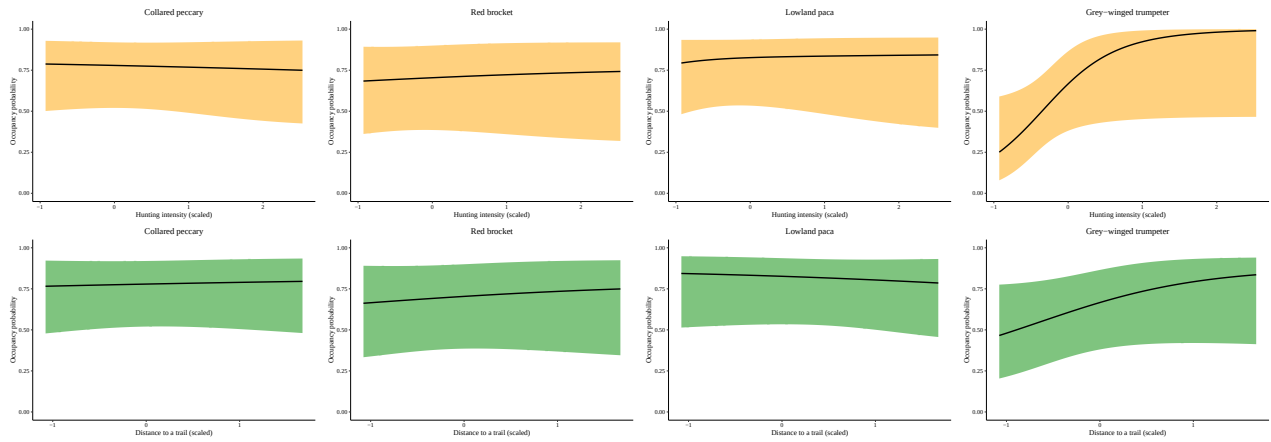
Community	nIndiv	nSpecies	shannonIndex	simpsonIndex
Siekopai	3201	44	2.232914	0.7679467
Sinangoe	2127	38	2.773124	0.9138674
Siona	762	35	2.529869	0.8475658
Zabalo	681	29	2.856870	0.9278184

Occupancy predictions

Sinangoe



Zabalo



Occupancy and detection estimates

Occupancy and detection are averaged across sites using actual covariate values as input.

Sinangoe

Table 12: Sinangoe occupancy and detection estimates

Species	avgOccupancy	avgOccupancySE	avgDetection	avgDetectionSE
Collared peccary	0.5198679	0.1114989	0.0854902	0.0152023
Red brocket	0.8667594	0.0722374	0.1605056	0.0138197
Lowland paca	0.7285123	0.0918352	0.9230655	0.0442927
Grey-winged trumpeter	0.5207976	0.1107963	0.1022458	0.0103941

Zabalo

Table 13: Zabalo occupancy and detection estimates

Species	avgOccupancy	avgOccupancySE	avgDetection	avgDetectionSE
Collared peccary	0.7774572	0.1222254	0.5010649	0.0654712
Red brocket	0.6980238	0.1685889	0.3416881	0.1150160
Lowland paca	0.7978465	0.1352453	0.0781317	0.0209527
Grey-winged trumpeter	0.5862868	0.1529942	0.0910051	0.0188624

